

SOLENZARA AB, France

WMO: 077650

Lat: 41.9239N

Lon: 9.3972E

Elev: 17

StdP: 101.12

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	1.8	3.2	-7.4	2.0	7.0	-5.4	2.4	7.4	11.2	11.5	10.0	11.8	3.5	270	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	31.2	22.3	30.2	22.5	29.3	22.3	24.8	29.3	24.1	28.7	23.4	28.2	3.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	18.0	28.5	22.4	17.2	28.1	21.7	16.4	27.5	75.2	29.4	72.4	28.8	69.9	28.2	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	7.8	6.5	0.0	34.9	1.4	1.6	-1.0	36.1	-1.9	37.0	-2.7	37.9	-3.7	39.1		
(5)			-1.9	25.6	1.3	0.8	-2.8	26.2	-3.6	26.6	-4.3	27.0	-5.2	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.6	9.8	9.7	11.7	14.2	17.7	21.7	24.6	24.8	21.7	18.0	13.6	10.8
	DBStd	5.91	2.45	2.53	2.38	2.13	2.05	2.46	1.94	1.79	2.19	2.42	2.75	2.72
	HDD10.0	102	32	32	11	1	0	0	0	0	0	0	5	22
	HDD18.3	1289	266	243	205	126	36	3	1	0	2	33	142	232
	CDD10.0	2503	26	22	64	126	240	351	454	460	349	249	114	48
	CDD18.3	648	0	0	1	1	17	105	196	202	102	24	1	0
	CDH23.3	5098	0	1	3	1	83	748	1825	1822	560	55	1	0
	CDH26.7	1331	0	0	1	0	4	141	549	558	76	3	0	0
Wind	WSAvg	3.0	3.1	3.2	3.3	3.0	2.8	2.8	2.9	2.9	3.1	2.9	3.0	3.2
Precipitation	PrecAvg	762	72	56	72	70	49	25	11	20	79	104	106	96
	PrecMax	1159	175	190	243	141	229	77	33	91	318	263	317	233
	PrecMin	524	0	7	0	12	8	1	0	0	14	2	20	0
	PrecStd	160	44	45	67	37	45	20	10	23	66	63	56	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.2	18.6	21.8	22.9	27.2	31.3	32.9	33.3	29.9	26.6	22.4	20.2
		MCWB	11.8	11.5	13.3	14.8	18.0	21.1	20.0	21.6	20.4	18.4	16.5	12.9
	2%	DB	16.7	16.7	18.9	21.3	25.6	29.6	31.2	31.3	28.4	24.8	20.4	17.9
		MCWB	11.0	10.6	12.0	14.7	17.6	21.9	22.5	22.5	20.9	19.1	15.5	12.5
	5%	DB	15.2	15.3	17.3	20.0	24.0	28.3	30.3	30.2	27.5	23.5	19.3	16.6
		MCWB	10.9	10.7	11.9	14.1	17.2	21.2	22.6	22.7	20.6	18.6	15.1	12.2
	10%	DB	14.1	14.4	16.3	18.9	22.7	27.1	29.4	29.3	26.5	22.4	18.2	15.3
		MCWB	10.5	10.3	11.8	13.9	16.9	20.6	22.5	22.4	20.0	17.9	14.5	11.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.7	13.4	14.8	17.1	20.7	24.2	25.7	25.6	24.1	21.7	18.6	15.2
		MCDB	16.1	15.9	18.5	20.8	24.9	29.0	30.3	30.0	27.7	24.3	20.4	17.5
	2%	WB	12.7	12.4	13.9	16.0	19.3	23.3	24.8	24.9	22.9	20.7	17.2	14.1
		MCDB	15.1	15.1	16.8	19.6	23.5	28.1	29.5	29.4	26.7	23.3	19.3	16.4
	5%	WB	11.9	11.7	13.2	15.3	18.5	22.2	24.1	24.2	22.0	19.7	16.1	13.1
		MCDB	14.3	14.4	16.0	18.7	22.6	27.0	28.9	28.8	26.2	22.5	18.3	15.5
	10%	WB	11.0	10.9	12.6	14.6	17.7	21.4	23.4	23.6	21.1	18.8	15.1	12.2
		MCDB	13.6	13.6	15.4	17.9	21.6	26.1	28.3	28.3	25.1	21.7	17.6	14.6
Mean Daily Temperature Range	5% DB	MDBR	7.1	7.9	8.1	8.3	8.6	9.0	9.0	8.8	8.3	7.8	7.3	7.0
		MCDBR	8.3	9.0	9.6	9.4	10.1	9.8	9.8	9.7	9.1	8.5	8.0	7.7
		MCWBR	5.9	6.3	6.3	6.0	6.3	6.3	6.7	7.1	6.3	5.9	5.3	5.5
	5% WB	MCDBR	7.1	8.1	8.4	8.4	8.9	8.9	8.9	8.9	8.3	7.4	6.9	6.9
		MCWBR	5.5	6.1	6.1	6.0	6.2	6.4	6.9	7.1	6.3	5.7	5.2	5.3
Clear-Sky Solar Irradiance	taub	0.333	0.351	0.382	0.420	0.432	0.420	0.420	0.417	0.409	0.389	0.366	0.336	
	taud	2.493	2.430	2.353	2.249	2.258	2.336	2.340	2.359	2.375	2.418	2.424	2.488	
	Ebn at Noon	812	851	862	850	847	857	852	844	823	796	759	774	
	Edh at Noon	75	92	111	132	134	125	123	117	108	92	80	70	
All-Sky Solar Radiation	RadAvg	1.75	2.64	3.89	5.19	6.24	7.10	7.23	6.28	4.72	3.20	1.97	1.56	
	RadStd	0.17	0.25	0.38	0.31	0.46	0.34	0.28	0.31	0.21	0.27	0.17	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)		+0.24	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+86	N/A

Nomenclature: See separate page